

Demand Velocity & Inventory Intelligence System

1. Project Overview

This project focuses on analyzing retail demand patterns and generating inventory intelligence for dark store operations using historical sales data.

The system is designed to identify demand fluctuations caused by:

- Historical demand movement
- Seasonal behavior
- Event-driven demand spikes
- Product demand velocity

The project uses the M5 Forecasting (Walmart) dataset as a retail simulation dataset to study product-level demand behavior and inventory movement.

The main objective of the project is to generate:

- Demand acceleration insights
 - Inventory risk alerts
 - Seasonal demand analysis
 - Event-driven demand interpretation
 - Replenishment recommendations
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2. Problem Statement

Traditional inventory systems mainly depend on static reorder rules and historical reports. These systems fail to properly respond to short-term demand fluctuations caused by seasonal patterns and special events.

This creates:

- Understocking during demand spikes
- Overstocking due to inaccurate inventory planning

The project aims to identify product-level demand movement and inventory risks using measurable historical and event-driven factors.

3. Project Objective

The objective of this project is to build a Demand Velocity and Inventory Intelligence Dashboard capable of:

- Tracking product demand movement
 - Identifying accelerating or declining products
 - Detecting seasonal and event-driven demand spikes
 - Generating inventory risk insights
 - Supporting replenishment planning
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4. Dataset Overview

The project uses the M5 Forecasting (Walmart) dataset.

The dataset contains:

- Historical daily sales data
- Product hierarchy information
- Store-level sales records
- Calendar and event information
- SNAP event indicators

The dataset is used as a retail simulation environment for studying inventory behavior and demand fluctuations.

5. Finalized Product Selection

The project focuses on 8 finalized products selected based on their fluctuation behavior and operational relevance.

Product	Purpose
Fresh Whole Turkey	Event-driven demand spikes
Halloween Candy	Seasonal event analysis
Fresh Strawberries	Perishable seasonal inventory
Vanilla Ice Cream	Seasonal demand acceleration
Hot Cocoa Mix	Winter seasonal demand
Tortilla Chips	Sports and event spikes
Fresh Whole Milk	Stable operational demand
Sliced White Bread	Daily replenishment baseline

The selected products represent:

- Stable demand products
- Seasonal products
- Event-driven products
- Perishable inventory products

This creates a balanced inventory intelligence system.

6. Finalized Parameters

The project uses four finalized parameter groups.

Historical Demand Parameters

Used for identifying previous sales behavior and demand trends.

Included features:

- Lag-1 Demand
 - Lag-7 Demand
 - 7-Day Rolling Average
 - 28-Day Rolling Average
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Seasonal Parameters

Used for identifying recurring calendar-based demand behavior.

Included features:

- Weekday
 - Weekend Flag
 - Month
 - Holiday Flag
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Event-Driven Parameters

Used for identifying demand spikes caused by events and holidays.

Included features:

- Event Name
 - Event Type
 - SNAP Day
 - Days to Event
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Demand Velocity Intelligence

Demand Velocity measures how fast product demand is growing or declining.

Velocity Formula:

Velocity Score =
$$\frac{((\text{Recent 28-Day Demand} - \text{Previous 28-Day Demand})}{\text{Previous 28-Day Demand}} \times 100$$

Velocity Categories:

Velocity Range	Status
< -50%	Critical Decline
-50% to -10%	Declining
-10% to +10%	Stable
+10% to +40%	Growing
> +40%	Accelerating

This parameter is used to identify rapidly growing or declining inventory movement.

7. Dashboard Architecture

The dashboard contains the following finalized panels.

Executive Overview Panel

Displays:

- Total predicted demand
 - High-risk products
 - Velocity summary
 - Active event indicators
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Product Velocity Intelligence Panel

Displays:

- Velocity Gauge
 - Product acceleration or decline status
 - Demand growth indicators
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Event Impact Panel

Displays:

- Event-driven demand spikes
 - Product-event relationships
 - Event influence analysis
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Seasonal Trend Panel

Displays:

- Monthly trends
 - Weekly patterns
 - Seasonal peaks and troughs
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Inventory Risk Intelligence Panel

Displays:

- Current inventory status
 - Predicted demand
 - Stockout risk
 - Overstock indication
 - Recommended replenishment quantity
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Product Deep Dive Panel

Displays:

- Historical demand trends
 - Velocity analysis
 - Seasonal behavior
 - Event influence analysis
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Explainability & Insight Panel

Displays:

- Reasons behind demand movement
 - Seasonal influence
 - Event contribution
 - Historical trend interpretation
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